ESG Reporting & AI/GenAI in Finance — Hands-On

Hands-On Complete-Knowledge Deep-Dive (India, 2026)

Do-the-task drills: worked examples, entries, interview drills & free practice

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The ESG Landscape & Career Map

What you'll be able to do

Explain, to a hiring panel or a portfolio manager, exactly what ESG is, *who pays for it and why*, and where a finance-trained person fits. You'll be able to name the roles (ESG reporting/disclosure, ESG research & ratings, sustainable finance/investing, climate risk, assurance), say what each does day-to-day, map the India and global demand drivers (SEBI, RBI, investors, lenders, EU import rules), and build a realistic 90-day plan to break in from an MBA-Finance + NISM-RA + trading background.

The essentials

ESG = Environmental, Social, Governance — a set of *non-financial* factors that are increasingly financial. It is not "CSR" (philanthropy) and not "sustainability" (a broader idea). ESG is specifically the lens investors, lenders and regulators use to price risk and allocate capital.

- **E** — GHG emissions, energy, water, waste, biodiversity, physical & transition climate risk.
- **S** — labour, health & safety, human rights, DEI, supply-chain, community, data privacy, customer welfare.
- **G** — board composition/independence, executive pay, audit quality, shareholder rights, bribery/corruption, related-party transactions.

The three demand drivers (memorise this triangle):

Driver	What they want	India instrument	Global instrument
Investors	Comparable, decision-useful data to price risk & screen	Institutional stewardship, ESG funds, proxy voting	ISSB S1/S2, PRI signatories, MSCI/Sustainalytics ratings
Regulators	Mandatory, assured disclosure	SEBI **BRSR** & **BRSR Core**, RBI climate risk guidance	EU **CSRD/ESRS**, ISSB adoption, SEC climate rule
Lenders / trade	Credit & counterparty risk; market access	RBI green deposit framework, sovereign green bonds	EU **CBAM** (carbon border tax), sustainability-linked loans

Two ideas that separate a novice from a pro: 1. **Materiality** — only ESG issues that affect the business (or that the business affects) matter. A cement firm's material issue is CO_2; a bank's is data privacy and financed emissions. "Financial materiality" (impact on enterprise value) vs "double materiality" (also impact *on the world*) is the single most-tested concept — see chapter 3. 2. **Greenwashing** — overstating green credentials. Regulators (SEBI, EU) now police it; assurance exists to prevent it.

The India timeline that matters (2026 reality): - **BRSR** mandatory for **top 1,000 listed companies by market cap** (since FY 2022-23). - **BRSR Core** — a subset of ~KPIs requiring **reasonable assurance** — phased in: top 150 (FY24), top 250 (FY25), top 500 (FY26), top 1,000 (FY27). Also extends to a listed company's **value chain** (top upstream/downstream partners) on a comply-or-explain basis. - SEBI **ESG rating providers (ERPs)** are now a *regulated* category (registration required).

Hands-on — step by step

Goal: map your own entry point. We'll carry a worked example: *you, targeting an ESG analyst role at a GCC (global capability centre) in Bengaluru or a Big-4 sustainability practice.*

Step 1 — Pick a lane. Score yourself (1-5) on fit:

Role	Core task	Your edge (trading + NISM + CA-inter)
ESG reporting/disclosure	Draft BRSR/CSRD reports, collect KPI data, run assurance	CA-inter (controls, assurance vocabulary) → **strong**
ESG research/ratings	Score companies, write issuer reports	NISM-RA + equity research instinct → **strong**
Sustainable finance / ESG investing	Screen funds, build ESG-tilted portfolios, green bonds	Trading desk + markets fluency → **strong**
Climate risk (bank/insurer)	Scenario analysis, financed emissions, stress tests	Finance + Python/Excel → **medium-strong**
ESG assurance (audit firm)	Limited/reasonable assurance engagements	CA-inter → **strong**

Step 2 — Learn the "spine" frameworks (chapters 2-4 do the deep work): BRSR (India), ISSB S1/S2 (global baseline), GHG Protocol (carbon), plus MSCI/Sustainalytics rating logic.

Step 3 — Build one portfolio artefact. Take a *real* Indian company's latest BRSR (all are public on the exchange/company site — e.g., a Tata or Infosys BRSR). Rebuild a slice: extract Scope 1/2 emissions, energy intensity, LTIFR (safety), board independence %, and write a one-page issuer note with an ESG opinion. This single artefact beats three certificates.

Step 4 — Certifications that actually help (in order of ROI): - **CFA Institute Certificate in ESG Investing** (~US$790; India's most-recognised ESG credential in finance). - **GARP SCR** (Sustainability & Climate Risk) — good for bank/climate-risk roles. - Free: **UN PRI** modules, **GHG Protocol** courses, SEBI/ICAI BRSR guidance.

Step 5 — Target the employers. - **GCCs / captives** (JPMorgan, Barclays, HSBC, Deloitte USI) — ESG data & reporting teams in India serving global parents. Highest-volume entry. - **Big-4 + consulting** (KPMG, EY, PwC, Deloitte) — BRSR advisory & assurance. - **Rating/data** (MSCI, Sustainalytics/Morningstar, S&P, CRISIL, ICRA ESG). - **Corporates** (Tata, Reliance, Infosys sustainability teams) and **AMCs** (ESG funds).

Step 6 — CV lines that land. Convert your background: "*Built issuer ESG note from BRSR Core KPIs; computed Scope 1+2 footprint (12,400 tCO_2e) and benchmarked energy intensity vs sector.*"

The output

A **one-page career map** you can act on:

> **Target role:** ESG Research Analyst (GCC, Bengaluru), backup: BRSR assurance associate (Big-4). **Spine skills:** BRSR/BRSR Core, ISSB S1/S2, GHG Protocol Scopes 1-3, MSCI/Sustainalytics scoring, double materiality. **Proof artefact:** 1-page issuer ESG note on [Company X] from its FY25 BRSR. **Credential:** CFA ESG Investing (in progress). **90-day plan:** Wk 1-3 frameworks → Wk 4-6 rebuild a BRSR slice → Wk 7-9 CFA-ESG + carbon calc → Wk 10-12 apply to GCCs/Big-4 with the artefact.

Checks, gotchas & red flags

- **Don't conflate CSR, sustainability, and ESG.** CSR (India: 2% spend under Companies Act §135) is philanthropy; ESG is risk/disclosure. Saying "ESG is CSR" in an interview is fatal.
- **ESG ≠ ethics-only.** Frame it as *risk and value*, not morality — that's how finance employers think.
- **Ratings disagree.** MSCI and Sustainalytics can rate the same firm very differently — know *why* (methodology, materiality weights). Chapter 4 covers this.
- **Greenwashing is now a compliance issue**, not PR. SEBI can act on misleading green claims.
- **Red flag in your own pitch:** vague passion ("I care about the planet") with no framework fluency. Employers hire framework + data skill, not enthusiasm.

Interview drill

Q1. "Why is ESG a finance topic, not a CSR topic?" *A:* Because it moves capital and prices risk. Investors use ISSB/ratings to screen; SEBI mandates assured BRSR disclosure; the EU's CBAM puts a cash cost on embedded carbon for exporters. ESG factors show up as cost of capital, credit spreads, insurance premiums and market access — CSR is discretionary spend, ESG is enterprise-value-relevant risk management.

Q2. "What's changed in Indian ESG regulation recently?" *A:* BRSR became mandatory for the top 1,000 listed firms, and **BRSR Core** introduced *reasonable assurance* on a defined KPI set, phased top-150 → top-1,000 by FY27, and extended to value chains on a comply-or-explain basis. SEBI also brought **ESG rating providers under registration**, reducing the wild-west problem in scoring.

Q3. "Financial vs double materiality — which does India use?" *A:* BRSR leans toward stakeholder/double-materiality framing (impact *of and on*), while ISSB S1/S2 is single/financial materiality (impact on enterprise value). The EU CSRD is explicitly double materiality. Knowing which lens a report uses tells you what a KPI is *for*.

Learn/practise (free)

- **SEBI BRSR circulars & ICAI's BRSR guidance** — the primary source; download the reporting format.
- **UN PRI "Introductory" modules** and **ISSB/IFRS Sustainability** free knowledge hub.
- **GHG Protocol** free courses and calculation tools (used in chapter 4).
- **Real BRSR reports** on any large-cap company's investor-relations page — rebuild a KPI slice as your portfolio artefact.
- **Practice rep:** each week, pick one listed company, pull its material ESG issues, and write a 200-word "ESG risk view" as if briefing a PM. Ten of these = interview-ready fluency.

BRSR (India) Hands-On

What you'll be able to do

Open SEBI's Business Responsibility and Sustainability Report (BRSR) format and actually *fill it in*: navigate its three sections, answer under the 9 NGRBC principles, compute the headline KPIs (emissions intensity, energy intensity, water, LTIFR, wage ratios, CSR, complaints), identify which KPIs sit inside **BRSR Core** and therefore need **reasonable assurance**, and produce a clean, assurance-ready sample disclosure. You'll know the applicability, timeline and value-chain reach cold.

The essentials

What BRSR is. A mandatory ESG disclosure filed as part of the annual report by the **top 1,000 listed companies by market capitalisation** (since FY 2022-23), replacing the old BRR. It is standardised so investors can compare.

Structure — three sections:

Section	Contents
A – General disclosures	Entity details, products/services, operations, employees & workers (gender split, differently-abled), CSR applicability, transparency/complaints
B – Management & process	Policies mapped to each of the 9 NGRBC principles, governance oversight, review frequency
C – Principle-wise performance	The quantitative meat: KPIs under each of the 9 principles, split into **Essential** (mandatory) and **Leadership** (voluntary/aspirational) indicators

The 9 NGRBC principles (National Guidelines on Responsible Business Conduct):

#	Principle (short)
P1	Ethics, transparency, accountability (anti-bribery, governance)
P2	Sustainable & safe goods/services (product lifecycle, R&D)
P3	Employee wellbeing & safety (wages, benefits, LTIFR)
P4	Stakeholder responsiveness
P5	Human rights
P6	Environment (energy, emissions, water, waste, biodiversity)
P7	Responsible public policy advocacy
P8	Inclusive growth & equitable development (CSR, social impact)
P9	Consumer value (product safety, data privacy, grievance)

BRSR Core = a focused set of **9 attributes / ~ KPIs** carved out for **reasonable assurance** (a higher bar than limited assurance). The nine attributes broadly cover: GHG intensity, water intensity, energy intensity, embracing circularity/waste, employee wellbeing & safety spend, gender diversity & wages, inclusive development/input sourcing from MSMEs/small producers, consumer complaints/data privacy, and openness of business (concentration of purchases/sales, related-party txns).

Assurance timeline (comply on a glide path):

Financial year	BRSR Core reasonable assurance applies to
FY 2023-24	Top 150 listed
FY 2024-25	Top 250
FY 2025-26	Top 500
FY 2026-27	Top 1,000

Value chain: top-250 firms disclose value-chain ESG (covering partners that are ~75% of purchases/sales) on a **comply-or-explain** basis, with assurance/assessment phased later.

Hands-on — step by step

Worked example: "Deccan Cements Ltd" (fictional), a top-250 firm, FY 2025-26. Data given: revenue ₹4,200 cr; Scope 1 = 9,800 tCO_2e; Scope 2 = 2,600 tCO_2e; total energy = 145,000 GJ; water withdrawn = 320,000 kL; 2 lost-time injuries; 1,800,000 hours worked; CSR obligation ₹6.4 cr, spent ₹6.5 cr.

Step 1 — Section A employee table. Report total employees (say 1,240: 1,090 male / 150 female) and workers separately, with differently-abled counts. *Gotcha:* "employees" (permanent) and "workers" (on-roll + contractual) are reported in *separate* tables — do not merge.

Step 2 — Section B policy map. For each of P1-P9 answer: policy exists? (Y/N), board-approved? (Y/N), weblink, translated to procedures?, review frequency. Deccan has policies for all 9; P7 (advocacy) marked "N/A — no formal advocacy."

Step 3 — Section C, P6 environment (the assured core). Compute the intensities:

- **GHG intensity** = (Scope 1 + Scope 2) / turnover = (9,800 + 2,600) / 4,200 = **2.95 tCO_2e per ₹ cr**. (Also report per unit of product for a cement firm.)
- **Energy intensity** = 145,000 GJ / 4,200 = **34.5 GJ per ₹ cr**; also show % from renewables (say 18%).
- **Water intensity** = 320,000 kL / 4,200 = **76.2 kL per ₹ cr**; disclose withdrawal by source and whether in water-stressed zones.

Step 4 — Section C, P3 safety. - **LTIFR (per million hours)** = (lost-time injuries × 1,000,000) / total hours = (2 × 1,000,000) / 1,800,000 = **1.11**. Report separately for employees and workers.

Step 5 — Section C, P5/P3 wages & diversity. Median remuneration, % female in workforce (150/1,240 = **12.1%**), gross wages paid to females as % of total wages, and minimum-wage compliance.

Step 6 — Section C, P8 CSR. CSR spent ₹6.5 cr vs obligation ₹6.4 cr → **101.6%** — compliant under Companies Act §135.

Step 7 — Section C, P9 consumer. Data-privacy policy status, number of consumer complaints (say 42 received, 40 resolved), instances of data breach (0).

Step 8 — Mark BRSR Core KPIs and route to assurance. GHG/energy/water intensity, LTIFR, wage/diversity, complaints, purchase/sales concentration → flag each with a tie-out to source data (meters, HR system, GL) so the **reasonable assurance** provider can vouch it. Build a KPI register: *KPI → formula → source document → owner → value*.

The output

A finished P6 disclosure block (assurance-ready):

> **Principle 6 — Environment (BRSR Core, reasonably assured)** | KPI | FY25-26 | FY24-25 | Unit | |---|---|---|---| | Total Scope 1 | 9,800 | 10,050 | tCO$_2$e | | Total Scope 2 | 2,600 | 2,780 | tCO$_2$e | | **GHG intensity** | **2.95** | 3.21 | tCO$_2$e / ₹ cr turnover | | Total energy consumed | 145,000 | 149,200 | GJ | | **Energy intensity** | **34.5** | 36.4 | GJ / ₹ cr | | Renewable energy share | 18% | 12% | % | | Water withdrawn | 320,000 | 335,000 | kL | | **Water intensity** | **76.2** | 79.8 | kL / ₹ cr |
>
> *Assurance:* Reasonable assurance obtained from [Assurance Provider LLP] per SEBI BRSR Core; assurance statement annexed. Prior-year comparatives restated for methodology consistency.

Checks, gotchas & red flags

- **Intensity denominators must be consistent** — turnover in the *same units and same currency* year over year; disclose the denominator explicitly. Mixing "per rupee crore" and "per unit output" without labels is the classic error.
- **Employees vs workers must not be combined** — SEBI treats them separately; assurors check this.
- **Scope 2 method** — state whether location-based or market-based (chapter 4). Auditors will ask.
- **Restatements** must be flagged and explained; silent changes are a red flag.
- **Reasonable ≠ limited assurance.** Reasonable is a *positive* opinion ("the KPI is fairly stated"); limited is *negative* ("nothing came to our attention"). BRSR Core demands reasonable — plan evidence accordingly.
- **Value-chain disclosures** are comply-or-explain — a blank with no explanation fails.
- **Assurance provider independence** — cannot be the same firm doing other conflicting work; SEBI restricts this.

Interview drill

Q1. "Which companies must get reasonable assurance on BRSR Core in FY 2025-26?" *A:* The **top 500 listed companies by market cap** in FY25-26, expanding to the top 1,000 in FY26-27. BRSR itself already applies to the top 1,000; Core assurance is the phased overlay. Value-chain ESG applies to the top 250 on a comply-or-explain basis.

Q2. "Compute GHG intensity from Scope 1 = 9,800, Scope 2 = 2,600, turnover ₹4,200 cr." *A:* (9,800 + 2,600)/4,200 = **2.95 tCO$_2$e per ₹ crore of turnover**. I'd also present it per unit of physical output since that's more decision-useful for an emissions-intensive manufacturer, and state whether Scope 2 is location- or market-based.

Q3. "What's the difference between BRSR and BRSR Core?" *A:* BRSR is the full nine-principle disclosure. BRSR Core is a defined subset of ~9 attributes/KPIs (intensities, safety, diversity, complaints, business openness) that SEBI singled out for **reasonable assurance** and value-chain reporting, precisely to give investors data they can trust and compare.

Learn/practise (free)

- **SEBI master circular on BRSR/BRSR Core** and the **ICAI Guidance Note on BRSR** — the exact format and Core KPI list.
- Download **two real BRSRs** (a manufacturer and a bank) from company IR pages; note how material issues differ.
- **Rehearse:** build a KPI register in Excel — column for KPI, formula, source doc, value — for one real company, then draft its P6 table. This *is* the entry-level job; doing it once makes you hireable.
- **NGRBC guidelines** (MCA) for the plain-English intent behind each principle.

Global Frameworks: GRI, SASB, TCFD, ISSB (S1/S2), EU CSRD/ESRS

What you'll be able to do

Tell any interviewer *which framework does what*, why there are several, and how they now consolidate. You'll define **financial vs double materiality** with a concrete example, recite **TCFD's 4 pillars**, explain **ISSB S1 and S2** as the emerging global baseline, scope **EU CSRD/ESRS** (who must report, when, and its double-materiality core), and **map all of them back to India's BRSR**. You'll run a small **materiality assessment** end-to-end.

The essentials

Think of the frameworks in two families — *impact-oriented* vs *investor/enterprise-value-oriented* — now converging under the ISSB.

Framework	Owner	Question it answers	Materiality	Status 2026
GRI	Global Reporting Initiative	"What is our impact **on the world**?"	Impact (outward)	Widely used, esp. EU; interoperable with ISSB
SASB	Now under ISSB/IFRS	"Which ESG issues are **financially material by industry**?"	Financial	Absorbed into ISSB; 77 industry standards live
TCFD	FSB task force	"How does **climate** affect our finances?"	Financial	Disbanded 2023; content **folded into ISSB S2**
ISSB S1/S2	IFRS Foundation	Global **baseline** for sustainability + climate to investors	Financial (enterprise value)	The emerging global standard; jurisdictions adopting
CSRD / ESRS	EU	Mandatory EU corporate sustainability reporting	**Double** materiality	In force, phased; deep and prescriptive
BRSR	SEBI (India)	India's mandatory listed-co disclosure	Stakeholder/double leaning	Mandatory top-1,000 (chapter 2)

Financial vs double materiality — the one concept to nail. - **Financial (single, "outside-in") materiality:** an ESG issue matters if it affects the company's cash flows, cost of capital or enterprise value. *Example:* a carbon price raises a steelmaker's costs → material. This is **ISSB's** lens. - **Double materiality:** report an issue if it is financially material **OR** if the company has a significant **impact on people/environment** ("inside-out"), even when it doesn't hit the P&L. *Example:* the steelmaker's local air pollution harms a community but doesn't (yet) cost the company — still reported. This is **CSRD/ESRS** and, in spirit, GRI and BRSR.

TCFD's 4 pillars (now the skeleton of ISSB S2, worth memorising): 1. **Governance** — board & management oversight of climate. 2. **Strategy** — climate risks/opportunities and their impact, incl. **scenario analysis** (e.g., 1.5°C, 2°C, >3°C). 3. **Risk management** — how climate risk is identified, assessed, integrated into ERM. 4.

Metrics & targets — GHG (Scope 1/2/3), internal carbon price, targets. Risk types: **physical** (acute events, chronic shifts) vs **transition** (policy, tech, market, reputation).

ISSB S1 & S2: - **IFRS S1** — general requirements: disclose *all* sustainability-related risks/opportunities that could affect enterprise value, using the TCFD four-pillar structure and SASB industry topics. - **IFRS S2** — climate-specific: builds TCFD in, **requires Scope 1, 2 and 3** GHG (with reliefs/phasing), industry metrics, and scenario analysis. ISSB is a **baseline** — jurisdictions (incl. considerations in India) can build on it; it's designed to be interoperable with GRI for the impact side.

CSRD/ESRS scope (why Indian analysts care): - Applies to large EU companies and listed SMEs, **and non-EU companies with substantial EU turnover** — so Indian exporters/subsidiaries can be pulled in. - Reports against **ESRS** (12 standards: 2 cross-cutting + 5 environmental incl. E1 Climate + 4 social + 1 governance). - **Double materiality assessment is mandatory** and must be evidenced. - Requires **limited assurance** (moving toward reasonable), digitally tagged.

Hands-on — step by step

Worked example: run a materiality assessment for "Krishna Textiles Ltd", an Indian apparel exporter to the EU.

Step 1 — Build the ESG issue longlist from sector references (SASB Apparel standard + GRI + ESRS topics): GHG emissions, water use, chemical/dye pollution, labour rights in supply chain, worker safety, product chemical safety, business ethics, data privacy.

Step 2 — Score each issue on two axes. - **Financial materiality (0-5):** impact on enterprise value. - **Impact materiality (0-5):** severity × likelihood of the company's impact on people/planet.

Issue	Financial	Impact	Double-material?
GHG emissions	4	4	**Yes**
Water & dye pollution	3	5	**Yes** (impact-driven)
Supply-chain labour rights	4	5	**Yes**
Product chemical safety	4	3	Yes
Data privacy	2	1	No
Business ethics	3	2	Borderline — monitor

Step 3 — Set thresholds. Rule: material if *either* axis ≥ 3 (double-materiality logic). Data privacy drops out; the rest stay.

Step 4 — Plot a materiality matrix (impact on x, financial on y). Top-right cluster (GHG, labour, water) = priority disclosures.

Step 5 — Map to frameworks & KPIs. - GHG → ISSB **S2** + ESRS **E1** + BRSR **P6** → KPI: Scope 1/2/3, intensity. - Water/pollution → GRI 303/303, ESRS **E2/E3** → KPI: water intensity, hazardous discharge. - Labour → GRI 400-series, ESRS **S2** (value-chain workers), BRSR **P3/P5** → KPI: supplier audits, LTIFR, living-wage coverage.

Step 6 — Cross-map to BRSR so one data-collection effort feeds all reports: build a crosswalk table (KPI → BRSR principle → ISSB/ESRS reference). This is the practical superpower — companies hate reporting the same number four ways.

The output

A **materiality statement + crosswalk** (deliverable):

> **Krishna Textiles — Material topics FY26 (double materiality)** Priority: (1) Supply-chain labour rights, (2) GHG emissions, (3) Water & chemical pollution. Secondary: product chemical safety, business ethics.
>
> | Material topic | ISSB | ESRS | GRI | BRSR | Lead KPI |
> |---|---|---|---|---|---|
> | GHG emissions | S2 | E1 | 305 | P6 | Scope 1/2/3, intensity |
> | Water & pollution | — | E2/E3 | 303/306 | P6 | Water intensity, discharge |
> | Labour rights | S1 | S2 | 408/409/414 | P3/P5 | Supplier audits, living-wage % |
> | Product safety | S1+SASB | — | 416 | P9 | Restricted-substance compliance |
>
> *Basis:* issue material where financial ≥3 OR impact ≥3; assessment evidenced per CSRD requirements.

Checks, gotchas & red flags

- **Don't call frameworks interchangeable.** GRI = impact, SASB/ISSB = financial, CSRD = both. Mixing them up signals a fake.
- **TCFD "still exists"** — technically disbanded; its content lives in **ISSB S2**. Say that, don't say TCFD is dead-and-gone.
- **Scope 3 under ISSB S2 is required** (with first-year and phasing reliefs) — a common half-truth is "S2 is only Scope 1 and 2."
- **CSRD reach is extraterritorial** — an Indian firm with big EU turnover can be caught; check the non-EU thresholds.
- **Double materiality needs evidence** under CSRD — a matrix with no methodology fails assurance.
- **Interoperability ≠ identical** — ISSB + GRI are designed to work together but have different objectives; use both, don't substitute.

Interview drill

Q1. "Explain double vs financial materiality with an example." *A:* Financial materiality (ISSB) asks whether an ESG issue affects enterprise value — a carbon price hitting a cement firm's costs. Double materiality (CSRD/ESRS) adds the *impact* dimension: report an issue if the company significantly affects people or planet even without a P&L hit — e.g., a factory's water pollution harming a community. India's BRSR leans double/stakeholder; ISSB is single/financial.

Q2. "What happened to TCFD and where did it go?" *A:* The TCFD was disbanded in 2023 after completing its mandate; its **four pillars — governance, strategy, risk management, metrics & targets — were absorbed into IFRS S2**. So when I build a climate disclosure I'm effectively still doing TCFD, just under the ISSB umbrella, now with mandatory Scope 3 and scenario analysis.

Q3. "A company reports under BRSR — why also care about ISSB and CSRD?" *A:* Investors want the ISSB global baseline for comparability, and EU-exposed Indian firms get pulled into CSRD's double-materiality regime. Smart companies do one materiality assessment and one data-collection effort, then map the KPIs across BRSR, ISSB S1/S2 and ESRS via a crosswalk — avoiding four separate reporting silos.

Learn/practise (free)

- **IFRS Sustainability (ISSB) knowledge hub** — free S1/S2 basis-for-conclusions and comparison to TCFD/SASB.
- **GRI Standards** (free download) and the **GRI-ISSB interoperability guidance**.
- **EFRAG ESRS** materials for CSRD; **TCFD final recommendations** PDF (still the best plain-English climate primer).
- **SASB Standards navigator** — look up your target company's industry standard (free).
- **Rehearse:** run the two-axis materiality scoring above for three companies in different sectors and build the crosswalk table each time — it's the exact deliverable ESG consultants sell.

Carbon Accounting & ESG Data/Ratings

What you'll be able to do

Build a company's **greenhouse-gas footprint from scratch** under the GHG Protocol: classify activities into **Scopes 1, 2 and 3**, apply **emission factors**, compute **location-based vs market-based** Scope 2, and total the inventory in **tCO_2e**. You'll also read the **ESG ratings landscape** (MSCI, Sustainalytics, CDP, S&P) — how each scores, why they disagree — and spot **greenwashing** and where **assurance** fits.

The essentials

GHG Protocol is the global carbon-accounting standard. Emissions are grouped into three scopes:

Scope	Definition	Examples
Scope 1	Direct emissions from owned/controlled sources	Company boilers, furnaces, owned vehicles, process CO_2 (e.g., cement calcination), refrigerant leaks
Scope 2	Indirect from **purchased energy**	Purchased electricity, steam, heating/cooling
Scope 3	All other **value-chain** indirect (15 categories)	Purchased goods, business travel, commuting, use of sold products, logistics, waste, investments (financed emissions)

Seven greenhouse gases, each converted to **CO_2-equivalent (CO_2e)** via its **Global Warming Potential (GWP)**: CO_2 (GWP 1), CH_4 (~28), N_2O (~265), plus HFCs/PFCs/SF_6/NF_3 (hundreds to tens of thousands). *tCO_2e = activity × emission factor × GWP.*

Emission factor (EF) = emissions per unit of activity. Sources: **IPCC**, **DEFRA/UK Gov** (widely used, free), **India's CEA** grid emission factor (for Scope 2 electricity — India ~0.7-0.8 tCO_2/MWh depending on year; always cite the CEA version used).

Scope 2 — two methods (report both): - **Location-based:** use the *grid average* EF (e.g., CEA national/regional factor). - **Market-based:** use EF of the *contracted* electricity (e.g., 0 for renewable PPAs backed by instruments/RECs). This is where renewable purchases show up.

ESG ratings landscape:

Provider	What it rates	Scale	Angle
MSCI ESG	Financially material ESG risk mgmt vs peers	**AAA-CCC**	Risk/opportunity, industry-relative
Sustainalytics (Morningstar)	**ESG Risk** = unmanaged risk to enterprise value	**0-40+**, *lower = better*	Absolute risk exposure
CDP	Disclosure on climate/water/forests	**A to D-**	Disclosure quality, not risk
S&P Global CSA	Corporate Sustainability Assessment (DJSI)	**0-100**	Questionnaire-driven

Key trap: **MSCI high letter = good; Sustainalytics high number = bad.** They also weight materiality differently, so the *same firm* can look strong on one and weak on the other — always ask *which methodology* and *what's material to that industry*.

Greenwashing — overstating green credentials: vague "eco-friendly" claims, cherry-picked metrics, Scope 3 omission, net-zero targets with no interim milestones, buying cheap offsets instead of cutting. **Assurance** (limited/reasonable, per ISAE 3000 / SEBI BRSR Core) is the antidote — an independent check that the numbers are fairly stated.

Hands-on — step by step

Worked example: "Deccan Cements Ltd" FY26 carbon footprint. Activity data: - Coal burnt in kiln: 4,000 tonnes; EF (coal) ≈ 2.42 tCO_2e per tonne. - Diesel in owned trucks: 120,000 litres; EF ≈ 0.00268 tCO_2e/litre. - Process CO_2 from limestone calcination: 3,100 tCO_2e (given, from clinker chemistry). - Purchased electricity: 3,500 MWh; CEA grid EF = 0.71 tCO_2/MWh; but 900 MWh sourced via a solar PPA (EF 0). - Business air travel: 250,000 passenger-km; EF ≈ 0.00015 tCO_2e/pkm. - Purchased clinker/goods (Scope 3 cat 1): 5,000 tCO_2e (from supplier data).

Step 1 — Scope 1 (direct). - Coal: 4,000 × 2.42 = **9,680 tCO_2e** - Diesel: 120,000 × 0.00268 = **321.6 tCO_2e** - Process CO_2: **3,100 tCO_2e** - **Scope 1 total = 9,680 + 321.6 + 3,100 = 13,101.6 tCO_2e**

Step 2 — Scope 2 (purchased electricity), both methods. - **Location-based:** 3,500 MWh × 0.71 = **2,485 tCO_2e** (grid average applied to all consumption). - **Market-based:** only non-renewable 2,600 MWh × 0.71 = **1,846 tCO_2e**; the 900 MWh solar PPA = 0. The 639 tCO_2e gap is the credit for renewable procurement.

Step 3 — Scope 3 (value chain). - Air travel (cat 6): 250,000 × 0.00015 = **37.5 tCO_2e** - Purchased goods (cat 1): **5,000 tCO_2e** - **Scope 3 (partial) = 5,037.5 tCO_2e** (full inventory would screen all 15 categories).

Step 4 — Total inventory.

Scope	tCO_2e (location-based S2)
Scope 1	13,101.6
Scope 2 (location)	2,485.0
Scope 3 (partial)	5,037.5
Total	**20,624.1**

Step 5 — Intensity metric (ties to BRSR): if turnover = ₹4,200 cr, Scope 1+2 intensity = (13,101.6 + 2,485)/4,200 = **3.71 tCO_2e per ₹ cr**. Per tonne of cement (say 500,000 t) = (15,586.6)/500,000 = **0.031 tCO_2e/tonne** — the decision-useful figure for a cement peer comparison.

Step 6 — Document EFs & sources so it survives assurance: a factor register listing every EF, its source (CEA v2025, DEFRA 2025, IPCC), and the activity-data source (fuel invoices, electricity bills, travel system).

The output

GHG inventory statement (assurance-ready):

> **Deccan Cements Ltd — GHG Inventory FY 2025-26** (GHG Protocol, operational control) | Scope | tCO_2e | Basis | |---|---|---| | Scope 1 — stationary + mobile + process | 13,101.6 | Fuel invoices; DEFRA/IPCC EFs | | Scope 2 — location-based | 2,485.0 | 3,500 MWh × CEA 0.71 | | Scope 2 — market-based | 1,846.0 | Net of 900 MWh solar PPA | | Scope 3 — cat 1 + cat 6 (partial) | 5,037.5 | Supplier + travel data | | **Total (location-based)** | **20,624.1** | | | Scope 1+2 intensity | 3.71 tCO_2e/₹ cr | vs 4.02 prior year |
>
> *Assurance:* reasonable assurance per SEBI BRSR Core / ISAE 3000. EF register annexed. Base year FY22 restated for the solar PPA.

Checks, gotchas & red flags

- **Don't double-count.** An emission is Scope 1 for the emitter and Scope 3 for the buyer — never both within one entity's inventory.
- **Report *both* Scope 2 methods.** Reporting only market-based hides grid reality; only location-based hides renewable effort. GHG Protocol wants both.
- **Cite the exact EF vintage** (CEA year, DEFRA year). A stale grid factor is the top assurance finding.
- **Scope 3 is usually the biggest scope** (often 70-90% of a footprint) — omitting it is the classic greenwash. ISSB S2 now requires it (with phasing).
- **Offsets ≠ reductions.** Netting gross emissions with offsets to claim "net zero" without cutting gross is a red flag; disclose gross and offsets separately.
- **Ratings ≠ performance.** A high MSCI rating measures *risk management relative to peers*, not low emissions. Don't equate a good rating with a green company.
- **Unit hygiene:** tonnes vs kg, MWh vs kWh, CO_2 vs CO_2e — one slip moves totals by 1000×.

Interview drill

Q1. "Classify: employee flights, the diesel in our delivery trucks, and our purchased electricity." *A:* Owned-truck diesel is **Scope 1** (direct, controlled source). Purchased electricity is **Scope 2** (indirect from bought energy). Employee flights are **Scope 3, category 6 (business travel)** — value-chain indirect. If we used a third-party logistics fleet instead of owned trucks, that diesel would shift to Scope 3 category 4.

Q2. "Compute Scope 2 location vs market-based: 3,500 MWh, grid EF 0.71, of which 900 MWh is a solar PPA." *A:* Location-based applies grid average to everything: 3,500 × 0.71 = **2,485 tCO_2e**. Market-based credits the contracted renewable: non-renewable 2,600 × 0.71 = **1,846 tCO_2e**, solar = 0. The 639 tCO_2e difference is the reported benefit of the PPA; GHG Protocol requires disclosing both.

Q3. "MSCI rates this firm AAA but Sustainalytics gives it a 32 — is that contradictory?" *A:* Not necessarily. MSCI is industry-relative letter grades where AAA = best-managed risk versus peers; Sustainalytics is an absolute *ESG Risk* score where **higher is worse**, so 32 (high) signals substantial unmanaged risk. They weight materiality differently and answer different questions — I'd read the methodology and the underlying material issues rather than treat the labels as comparable.

Learn/practise (free)

- **GHG Protocol** — free Corporate Standard, Scope 2 Guidance, Scope 3 Standard, and Excel calculation tools.
- **DEFRA/UK Government conversion factors** (annual, free) and **IPCC** EFs; **India CEA CO_2 Baseline Database** for the grid factor.
- **CDP guidance** and **MSCI/Sustainalytics public methodology** documents — read one of each to see why scores diverge.
- **Rehearse:** take any company's fuel and electricity data from its BRSR/annual report and rebuild the Scope 1+2 inventory in Excel with an EF register, then compute intensity. Do it for a manufacturer and a services firm to feel how the scope mix flips (Scope 3 dominates services via purchased goods & travel).

GenAI for Finance Workflows: Copilots, LLMs & Prompting

What you'll be able to do

By the end of this chapter you can sit at a real finance desk and use generative AI as a competent junior analyst: turn a Copilot in Excel/Power BI into a formula-and-DAX author, draft month-end variance commentary from a numbers table, summarise a 40-page annual report or RBI circular into a decision-ready brief, generate and debug SQL/DAX/VBA you can read and trust, clean messy vendor master data, and — critically — *verify* every output before it leaves your hands. You will know exactly which finance tasks GenAI accelerates safely, which it must never do unsupervised, and the prompt patterns that get first-time-right results.

The essentials

A large language model (LLM) predicts the next token from your prompt plus its training data. It is a **language** engine, not a **calculation** engine: it drafts, summarises, translates, and codes brilliantly, but its arithmetic on free text is unreliable. So the rule for finance is: **let the LLM write the logic (formula, query, commentary), let a deterministic tool (Excel, SQL engine, calculator) do the maths.**

Tools you'll actually meet in an India GCC / analyst seat (mid-2026):

Tool	What it does	Access / cost (2026)
Microsoft 365 Copilot	Sits in Excel, Word, Outlook, Teams, PowerPoint; drafts, summarises emails/meetings, writes Excel formulas & pivot suggestions	Paid add-on, ~US$30/user/mo on top of a M365 licence; enterprise-provisioned
Copilot in Excel	Natural-language → formulas, PivotTables, conditional formatting, "explain this workbook", basic Python-in-Excel	Part of M365 Copilot; needs data in a Table
Copilot in Power BI / Fabric	Generate DAX measures, "create a report page for revenue by region", narrative summaries of visuals	Fabric capacity (F-SKU) or PPU licence
ChatGPT / Claude / Gemini (enterprise)	General research, drafting, code, data cleaning via chat or API	Enterprise tiers keep your data out of training; free tiers do NOT
Copilot free / Bing	The free way to practise all prompt patterns below	Free with a Microsoft account

Data-privacy non-negotiable: never paste unpublished financials, PII, MNPI (unpublished price-sensitive information under SEBI PIT Regulations), or client data into a *consumer* GenAI tool. Use only the enterprise instance your employer sanctions, where data isn't retained for training. When practising at home, use synthetic or already-public numbers.

Hands-on — step by step

Worked scenario: you're closing **June 2026** for a mid-cap. Actual revenue ₹1,240 lakh vs budget ₹1,150 lakh; actual opex ₹910 lakh vs budget ₹860 lakh.

1. Copilot in Excel — write a formula. Put your data in a Table (Ctrl+T). Open the Copilot pane (Home ribbon → Copilot). Prompt:

> "Add a column 'Variance %' = (Actual − Budget) / Budget, formatted as a percentage with one decimal, and flag values worse than −5% for expenses as 'Investigate'."

Copilot returns something like `=(B2-C2)/C2` and offers to insert the column. **Read the formula** — is Budget the denominator, sign convention correct? Insert, then eyeball one row by hand: (1240−1150)/1150 = 7.8%. Ties out.

2. LLM for variance commentary. Paste the *computed* variance table (numbers already correct in Excel) into your enterprise LLM:

> "You are an FP&A analyst. Below is June-26 actual vs budget (₹ lakh). Write 4 bullet points of variance commentary for the CFO: state the driver direction, quantify in ₹ lakh and %, and flag anything needing action. Neutral, factual tone. Do not invent causes — where a cause isn't given, say 'driver to be confirmed with business.'" Revenue: Actual 1240, Budget 1150. Opex: Actual 910, Budget 860.

The "do not invent causes" clause is what stops hallucinated narratives.

3. Draft DAX / SQL / VBA. In Power BI Copilot:

> "Create a DAX measure 'Rev Var %' = DIVIDE(SUM(Actual[Rev]) − SUM(Budget[Rev]), SUM(Budget[Rev])). Handle divide-by-zero."

You get `Rev Var % = DIVIDE(SUM(…)-SUM(…), SUM(…), BLANK())`. For SQL, prompt a chat LLM: "*Write ANSI SQL to return top 10 customers by FY26 revenue from table sales(cust_id, invoice_dt, amount); FY = Apr–Mar.*" Then **run it against a test set** before production.

4. Data cleaning. Paste a messy vendor list:

> "Standardise these vendor names to Title Case, strip 'Pvt Ltd/Private Limited' variants to 'Pvt Ltd', and list suspected duplicates. Return a two-column table: original | cleaned."

Use the mapping to build a lookup — never let the LLM silently overwrite a master.

Prompt patterns worth memorising (RTCF): Role ("You are a treasury analyst"), **T**ask (one verb, one deliverable), **C**ontext (the data, the accounting standard, the currency), **F**ormat ("as a 3-column markdown table"). Add **constraints** ("cite the row", "don't invent numbers", "flag uncertainty").

The output

The June commentary Copilot/LLM produces, after your review:

> **June-2026 Actual vs Budget — CFO note (₹ lakh)** - **Revenue ₹1,240 vs ₹1,150 budget: +₹90 (+7.8%)** — favourable; driver to be confirmed with business (volume vs price split not provided). - **Opex ₹910 vs ₹860 budget: +₹50 (+5.8%) adverse** — spend above plan; exceeds the −5% expense threshold, flagged for investigation. - **Net contribution ₹330 vs ₹290 budget: +₹40 (+13.8%)** — revenue upside outweighs cost overrun. - **Action:** confirm revenue driver and the ₹50 lakh opex overrun with cost-centre owners before sign-off.

Deliverable = a clean Excel table with a correct, self-checking `Variance %` column, a reusable DAX measure, and a CFO-ready paragraph — drafted in minutes, verified by you.

Checks, gotchas & red flags

- **Never trust LLM arithmetic on free text.** It will confidently return 7.2% where Excel says 7.8%. Compute in the sheet; use the LLM only to *narrate* computed numbers.
- **Hallucinated citations / causes.** LLMs invent plausible drivers ("higher marketing spend") that never happened. Force "driver to be confirmed" where data is silent.
- **Stale knowledge.** A model's training cutoff means it may quote the *old* tax slab, an outdated Ind AS, or a superseded RBI limit. Verify anything rules-based against the live source (incometax.gov.in, RBI, ICAI).
- **Privacy breach.** Pasting MNPI or client PII into a consumer tool can be a SEBI/DPDP Act violation and grounds for dismissal. Enterprise instance only.
- **Silent overwrites in data cleaning.** Always keep original + cleaned side by side; a "helpful" standardisation can merge two genuinely different vendors.
- **Copilot needs structured input.** Data must be in a proper Excel Table or a clean model; on merged cells and blank rows it fails or guesses.
- **Explainability.** If you can't read the DAX/SQL it wrote, you can't own it. Ask it to "explain this line by line" until you can.

Interview drill

Q1: "Your manager says 'just have ChatGPT write our quarterly variance report.' What do you do?" A: I'd use GenAI as a drafting assistant, not the author of record. I compute all variances in the model so the maths is deterministic, feed the *verified* numbers to an enterprise LLM with a role/format prompt and an explicit "don't invent drivers" constraint, then review every line against source, add the real business drivers from cost-centre owners, and take ownership before it goes to the CFO. I'd also confirm we're on the sanctioned enterprise tenant so no unpublished financials train a public model.

Q2: "Where does an LLM genuinely save an analyst time, and where is it dangerous?" A: Big wins: drafting commentary and emails, summarising long filings/circulars, writing first-draft SQL/DAX/VBA, and cleaning/standardising text data. Dangerous: doing the arithmetic, applying tax/accounting rules from memory (training cutoff risk), and anything touching MNPI or PII in a non-enterprise tool. The pattern is language-tasks yes, calculation-and-compliance-of-record no.

Q3: "How do you stop hallucinations in a finance workflow?" A: Ground it — give the model the actual data/document rather than asking from memory (RAG), constrain the output ("cite the row/section", "say 'unknown' if not present"), keep computation in a deterministic engine, and always human-verify against the primary source before use.

Learn/practise (free)

- **Copilot free / Microsoft Copilot** (copilot.microsoft.com) and **Excel's built-in "Analyze Data"** — rehearse formula and pivot prompts with public data.
- **ChatGPT / Claude / Gemini free tiers** — practise RTCF prompts using *synthetic* numbers only.
- **DAX:** Microsoft Learn "Copilot in Power BI" module + SQLBI free articles; verify every generated measure in Power BI Desktop (free).
- **Prompting:** Microsoft "Copilot Lab" prompt gallery, Anthropic's free prompt-engineering guide, Google's "Prompting essentials".
- **Rehearsal drill:** take any listed company's annual report (public), ask the LLM for a 10-bullet summary, then fact-check each bullet against the PDF — this trains both prompting and the verification reflex employers want.

AI Agents, RAG & Governance in Finance

What you'll be able to do

You'll be able to explain and reason about the three things a 2026 finance employer means when they say "we're deploying AI": **agentic automation** (AI that takes multi-step actions in finance operations), **RAG** (retrieval-augmented generation — an LLM answering strictly from *your* documents and policies), and **AI governance** (model risk, hallucination control, data privacy, and regulatory compliance including the EU AI Act). You'll be able to sketch how a RAG assistant over a policy library actually works, describe a realistic agentic invoice-to-pay or reconciliation workflow, name what leading firms (Genpact, Accenture, Citi) are actually doing, and — the part that gets you hired into a GCC or compliance/model-risk seat — articulate the *controls* that make any of this safe.

The essentials

RAG in one line: instead of relying on what the model memorised, you retrieve the relevant chunks from *your* trusted documents and paste them into the prompt, so the answer is grounded in — and citable to — your data.

The RAG pipeline:

Stage	What happens
Ingest	PDFs/policies/contracts split into chunks (~300–800 tokens)
Embed	Each chunk → a vector (embedding) capturing meaning
Store	Vectors go into a vector DB (e.g. Azure AI Search, Pinecone, pgvector)
Retrieve	User question is embedded; nearest chunks are fetched
Generate	LLM answers using *only* those chunks, with citations

RAG is how you get "Which vendors have net-45 payment terms per our contracts?" answered from the actual contracts, with source links — and how you *stop* hallucination, because the model is told "answer only from the provided context; if not present, say you don't know."

AI agents go one step further: an LLM that can *plan* and *call tools* (query a DB, post to an API, send an email, raise a ticket) in a loop until a goal is met. In finance ops this looks like:

- **Invoice-to-pay:** agent reads an invoice, extracts fields, matches to PO and GRN (3-way match), flags mismatches, drafts the posting — human approves.
- **Reconciliation:** agent pulls bank and ledger, matches transactions, proposes journal entries for breaks.
- **Close support:** agent gathers variance data, drafts commentary, chases open items.

Crucially, safe agents run **human-in-the-loop** for anything that moves money or hits the ledger, and with **least-privilege** access (read-mostly, scoped credentials).

Where the industry is (2026): - **Genpact** — "agentic AI" for finance & accounting BPO; agents for invoice processing, collections, and close, sold as outcome-based services. - **Accenture** — reinvention services + a large trained AI workforce; RAG "knowledge assistants" over client policy and control libraries. - **Citi / global**

banks — internal GenAI assistants (document search, coding, research summarisation) rolled out to tens of thousands of staff, kept inside a governed, private environment; heavy model-risk oversight.

Hands-on — step by step

You can't run a bank's stack at home, but you can build the exact pattern. Scenario: a **RAG assistant over your company's Travel & Expense policy** so employees stop emailing finance "can I claim airport lounge access?"

1. **Collect the source** — the T&E policy PDF (public/synthetic for practice).
2. **Chunk it** — split by section/heading into ~500-token pieces; keep the section title with each chunk (helps citation).
3. **Embed** — run each chunk through an embedding model (e.g. Azure OpenAI `text-embedding-3-small`, or a free local model via `sentence-transformers`).
4. **Store** — load vectors into a vector store. For practice, `pgvector` on free Postgres, or an in-memory FAISS index in Python.
5. **Query** — embed the question "Is airport lounge access reimbursable?", retrieve top-4 chunks.
6. **Generate with guardrails** — prompt:

> "Answer the employee's question using ONLY the policy sections below. Quote the clause and its section number. If the policy does not cover it, reply: 'Not specified in policy — check with Finance.' Do not guess."

7. **Show citations** — return the answer plus the section it came from, so a human can verify.

For the **agentic** flavour, add a tool: if the answer is "reimbursable", the agent can *draft* (not submit) a claim in the expense system, leaving final submission to the employee. That "draft, don't submit" boundary is the governance point.

The output

The RAG assistant's response, as a user sees it:

> **Q: Is airport lounge access reimbursable? A:** Yes, for international travel in Manager grade and above. Per **T&E Policy §4.3 (Airport Services):** "Lounge access is reimbursable for international itineraries for employees at M3 and above, capped at ₹2,500 per visit." For domestic travel it is **not** covered (§4.3, same clause). *Source: T&E Policy v2026.1, §4.3.*

And a one-page **governance register entry** for that assistant:

> **Use case:** T&E policy Q&A (RAG). **Risk tier:** Low (informational, no payment action). **Data:** internal policy only, no PII. **Controls:** answers cited to source; "not specified" fallback; enterprise LLM (no training on our data); quarterly review of accuracy on a 20-question test set; human owner: FP&A lead. **EU AI Act relevance:** minimal-risk / limited-risk (transparency: users told it's AI).

Checks, gotchas & red flags

- **RAG still hallucinates if retrieval is bad.** If the right chunk isn't retrieved, the model may fill the gap. Test retrieval quality, not just the final answer.
- **Chunking destroys tables.** Financial policies and filings have tables; naive splitting breaks them. Use layout-aware parsing for numbers.
- **Agents with write access are the real danger.** An agent that can post journals or release payments unsupervised is a fraud-and-error surface. Enforce human approval, maker-checker, and least privilege.
- **Data privacy (India DPDP Act 2023 + SEBI/RBI).** Personal data and MNPI must stay in a governed environment; know where the vector store and the LLM physically process data (data residency).
- **Model risk (RBI/SEBI + global norms).** Any model influencing financial decisions needs validation, monitoring, documentation, and an accountable owner — the same discipline as credit/market-risk models (think SR 11-7 / RBI model-risk expectations).
- **EU AI Act awareness.** Risk-tiered: *unacceptable* uses banned; *high-risk* (e.g. creditworthiness scoring) faces strict obligations — data governance, human oversight, transparency, logging; *limited-risk* needs disclosure that users are dealing with AI. GPAI (foundation-model) obligations phase in through 2025–2027. Even Indian GCCs serving EU clients must comply.
- **"Black box" excuse doesn't fly.** If you can't explain how the assistant reached an answer, it fails audit. Citations and logs are your defence.

Interview drill

Q1: "What is RAG and why does finance care?" A: Retrieval-augmented generation grounds an LLM in our own trusted documents: we embed our policies/filings, retrieve the relevant chunks for a question, and make the model answer only from those, with citations. Finance cares because it turns a hallucination-prone chatbot into an auditable assistant — every answer traces to a source clause, and if the source is silent the model says "unknown" instead of inventing. That auditability is what makes it usable under model-risk and compliance requirements.

Q2: "Where would you allow an AI agent to act autonomously in finance ops, and where not?" A: Autonomous is fine for *read and draft* work — extracting invoice fields, matching to PO/GRN, proposing journal entries, drafting commentary — because a human still approves. I would not let an agent autonomously post to the ledger, release payments, or change master data; those need maker-checker and least-privilege credentials. The principle: agents accelerate the preparation, humans retain the control point wherever money or the books move.

Q3: "A regulator asks how you control AI risk. What do you say?" A: We tier use cases by risk, keep an inventory with an accountable owner each, validate and monitor any model influencing decisions, ground answers via RAG with citations, log inputs/outputs, keep data in a governed residency-compliant environment (DPDP/RBI/SEBI), enforce human oversight on high-risk actions, and map obligations to the EU AI Act where we serve EU clients. It's the model-risk-management discipline extended to GenAI.

Learn/practise (free)

- **Build a RAG demo free:** Python + `sentence-transformers` + FAISS, or LangChain/LlamaIndex quick-starts; Azure has a free-tier "chat over your data" sample.

- **Microsoft Learn** — "Fundamentals of Generative AI", "Implement RAG with Azure AI Search" (free modules).
- **EU AI Act** — the official EU "AI Act Explorer" / summary pages; read the risk-tier definitions and the GPAI section.
- **Model risk** — read a public summary of Fed SR 11-7 and RBI's model-governance guidance to see the control vocabulary.
- **Vendor viewpoints** — Genpact, Accenture, and BCG public "agentic AI in finance" reports (free) for real use-case framing to cite in interviews.
- **Rehearsal:** build the T&E RAG bot on a public policy PDF, deliberately ask an out-of-scope question, and confirm it says "not specified" — that demo alone is strong interview evidence.

ESG & AI Interview Drills + Cheat-Sheet

What you'll be able to do

You'll be able to put your ESG and AI knowledge on a resume **honestly** (no inflated "AI/ML expert" claims that collapse in five minutes), and answer the exact ESG/AI questions an India analyst / GCC / compliance / treasury interviewer throws in 2026: explain BRSR and its Core, define Scope 1/2/3 with a worked example, describe a concrete finance AI use case end-to-end, and enumerate AI risks with controls. You'll walk out with a one-page cheat-sheet — frameworks, emission scopes, tools, and certifications — you can revise the night before.

The essentials

Positioning honestly. Match the claim to the evidence:

If you did...	Say this	Don't say
Read BRSR, built a disclosure map	"Familiar with BRSR structure and BRSR Core KPIs"	"Led ESG reporting"
Built a RAG demo / used Copilot	"Built a RAG proof-of-concept; use Copilot for FP&A drafting"	"AI/ML engineer"
Computed a carbon estimate in Excel	"Can compute Scope 1/2 from activity data"	"Carbon accounting specialist"

Interviewers respect a precise junior claim over a vague senior one. State the tool, the task, the dataset, and that you verify outputs.

ESG core facts (India, 2026): - **BRSR** = Business Responsibility & Sustainability Report; mandatory for the **top 1,000 listed companies by market cap** (SEBI), filed with the annual report. Structured on the **9 NGRBC principles**. - **BRSR Core** = a subset of key ESG KPIs with **reasonable assurance** required, being phased by market-cap tranche; includes **value-chain** disclosures (phased, comply-or-explain). - **Greenhouse-gas Scopes (GHG Protocol):** - **Scope 1** — direct emissions you own/control (company boilers, vehicles, generators). - **Scope 2** — indirect from **purchased** electricity/heat/steam. - **Scope 3** — all other value-chain emissions (purchased goods, business travel, product use, investments) — usually the largest and hardest. - **Global frameworks: GRI** (impact/stakeholder view), **ISSB IFRS S1 (general) & S2 (climate)** — the converging global baseline, **TCFD** (folded into ISSB), **CDP** (disclosure platform), **SASB** (industry metrics, now under ISSB), **EU CSRD/ESRS** (EU's mandatory regime — relevant to Indian firms in EU supply chains).

AI facts: LLM = language engine, not calculator; **RAG** grounds answers in your documents with citations; **agents** plan and call tools (keep humans on money-moving steps); **governance** = model risk, hallucination control, data privacy (DPDP Act), and the **EU AI Act** risk tiers (unacceptable / high / limited / minimal).

Hands-on — step by step (a worked Scope 1+2 calc)

A small unit's FY26 activity data: - Diesel in generators: **10,000 litres** - Company cars petrol: **5,000 litres** - Grid electricity: **200,000 kWh**

Emission factors (illustrative — always use the current CEA/India GHG Program / DEFRA factor): - Diesel ≈ **2.68 kg CO_2e / litre** - Petrol ≈ **2.31 kg CO_2e / litre** - India grid ≈ **0.71 kg CO_2e / kWh** (CEA average — verify the year's value)

Scope 1 (direct combustion): - Diesel: 10,000 × 2.68 = **26,800 kg** - Petrol: 5,000 × 2.31 = **11,550 kg** - Scope 1 = 26,800 + 11,550 = **38,350 kg = 38.35 tCO_2e**

Scope 2 (purchased electricity, location-based): - 200,000 × 0.71 = **142,000 kg = 142.0 tCO_2e**

Total Scope 1+2 = 180.35 tCO_2e. Scope 3 (say business travel, purchased goods) would be estimated separately and typically dwarfs this.

In Excel: one column of activity data × a factor column, `=SUMPRODUCT(activity, factor)/1000` to get tonnes. That's the whole mechanic — the difficulty is sourcing correct factors and complete activity data, and getting assurance-ready evidence.

The output

A resume ESG/AI block that survives scrutiny:

> **ESG & AI (finance):** Familiar with SEBI BRSR structure and BRSR Core KPIs; can compute Scope 1 & 2 emissions from activity data in Excel and map disclosures to GHG Protocol / ISSB S2. Working knowledge of GenAI for FP&A — Copilot-assisted variance commentary and DAX; built a RAG proof-of-concept answering questions over a policy PDF with source citations. Aware of AI-governance and EU AI Act risk tiers; verify all AI outputs against source.

And the interview cheat-sheet (below) folded into your revision.

Cheat-sheet

ESG frameworks

Framework	Owner	Point
BRSR / BRSR Core	SEBI	India mandatory (top 1,000); Core = assured KPIs + value chain
NGRBC (9 principles)	MCA	The backbone BRSR maps to
GRI	GRI	Impact on stakeholders/environment
ISSB IFRS S1/S2	IFRS Foundation	Global baseline; S2 = climate (absorbs TCFD)
CSRD / ESRS	EU	EU mandatory; double materiality
CDP	CDP	Disclosure platform (climate/water/forests)

GHG Scopes: 1 = own combustion; 2 = bought energy; 3 = value chain (up + downstream).

AI toolkit: LLM (draft/summarise, not calculate) · Copilot (Excel/Power BI/M365) · RAG (grounded, cited answers) · Agents (plan + act, human-in-loop) · Vector DB (embeddings store).

AI risks → controls: hallucination → RAG + citations + verify · privacy (DPDP/MNPI) → enterprise instance + residency · model risk → validation/monitoring/owner · autonomy → maker-checker + least privilege · regulation → EU AI Act risk tiers.

Certifications worth naming: GARP **SCR** (Sustainability & Climate Risk); CFA Institute **Certificate in ESG Investing**; **GRI Certified**; for AI, Microsoft **AI-900 / DP-900**, and vendor GenAI badges. Your **NISM RA** and CA-inter already signal core finance rigour.

Checks, gotchas & red flags

- **Wrong emission factor / year.** Factors change annually and by region; a stale India grid factor is the classic error. Cite the source and year.
- **Unit slips.** kg vs tonnes (÷1000), kWh vs MWh — mislabelled units are the most common Scope-2 mistake.
- **Scope 2 double-count / method.** Location-based vs market-based are different methods; don't mix them.
- **Over-claiming AI skill.** Saying "ML engineer" invites a coding grill you'll fail. Claim exactly what you built.
- **BRSR scope error.** It's the top 1,000 *listed* firms — not all companies, not by turnover. Know Core's assurance and value-chain phasing.
- **Confusing ISSB/TCFD/GRI.** ISSB serves investors (enterprise value); GRI serves broader stakeholders (impact). Different lenses, not rivals.

Interview drill

Q1: "Explain BRSR and BRSR Core." A: BRSR is SEBI's mandatory Business Responsibility & Sustainability Report for the top 1,000 listed companies, filed with the annual report and structured on the nine NGRBC principles covering ESG performance. BRSR Core is a defined subset of key ESG KPIs that require *reasonable assurance*, phased in by market-cap tranche, and it extends to value-chain disclosures on a comply-or-explain basis. The intent is comparable, assured, decision-useful ESG data rather than narrative-only reporting.

Q2: "What's the difference between Scope 1, 2 and 3? Give an example." A: Scope 1 is direct emissions from sources you own or control — say diesel burned in our generators. Scope 2 is indirect emissions from energy we purchase — our grid electricity. Scope 3 is everything else in the value chain, upstream and downstream — purchased goods, business travel, product use, investments. For our unit, 10,000 L diesel at ~2.68 kg/L is ~27 tonnes of Scope 1; 200,000 kWh at ~0.71 kg/kWh is ~142 tonnes of Scope 2. Scope 3 is usually the biggest and hardest to measure because it depends on suppliers and customers.

Q3: "Give me one AI use case in finance and its main risks." A: A RAG assistant over our policy library — employees ask "is lounge access reimbursable?" and it answers only from the policy with the clause cited. Risks: hallucination if retrieval misses the right section (mitigate with citation + a "not specified" fallback and accuracy testing), data privacy since policies may touch personal data (keep it on an enterprise, data-resident instance under the DPDP Act), and governance — an owner, monitoring, and logs so it passes audit. If we ever let it *act*, like drafting a claim, the human still submits — no autonomous money movement.

Learn/practise (free)

- **BRSR:** read a real filing — download any top-500 company's BRSR from BSE/NSE and trace the 9 principles and Core KPIs.
- **GHG Protocol:** the free Corporate Standard PDF + free GHG calculation tools; India GHG Program / CEA CO_2 baseline database for factors.

- **ISSB:** IFRS Foundation's free IFRS S1/S2 overview pages.
- **EU AI Act:** the EU "AI Act Explorer" (free) for risk tiers.
- **AI:** Microsoft Learn AI-900 path (free); build the policy-RAG demo from Chapter 6.
- **Rehearsal:** record yourself answering the three drills in under 90 seconds each; do the Scope 1+2 Excel calc from scratch until units and factors are automatic. That combination — a clean carbon calc plus an honest AI story with a working demo — is exactly what differentiates a credible 2026 finance candidate.